

# Multi-Task Learning Based Underwater Acoustic OFDM Communications

Hao Zhao<sup>†</sup>, Fei Ji<sup>†\*</sup>, Miaowen Wen<sup>†</sup>, Hua Yu<sup>†</sup> and Quansheng Guan<sup>†</sup>

<sup>†</sup>School of Electronic and Information Engineering, South China University of Technology

\*Key Laboratory of Marine Environmental Survey Technology and Application, Ministry of Natural Resources  
Guangzhou 510300, China

Email: ctzhaohao@mail.scut.edu.cn, {eefeiji, eemwwen, yuhua, eeqshguan}@scut.edu.cn

**Abstract**—Deep learning (DL) based receivers are trained under simulation dataset and tested in the real world. Unfortunately, there is no accurate channel model in underwater acoustic (UWA) communications. The mismatch between the training data set and the test environment degrades the receiver performance for UWA communications. Considering that the classical receiver contains channel estimation, equalization and demodulation tasks that are mutually dependent, in this paper, we propose a novel receiver, called TaskNet, based on multi-task learning (MTL) to improve the DL-receiver's generalization performance. To demonstrate the advantages of TaskNet, we establish a baseband UWA channel impulse response (CIR) dataset, which includes both simulated and measured CIRs, and make comparison with the classical single-task based DL-receivers including FC-DNN and ComNet. Through analysis and simulation results, the proposed MTL based DL-receiver is proved to have a better generalization performance than the single-task based ones, whether the signal suffers nonlinear damage, or the training set does not match the real channel.

**Index Terms**—Multi-task learning, underwater acoustic (UWA) communications, generalization, dataset.

## I. INTRODUCTION

Orthogonal frequency division multiplexing (OFDM) has prevailed in recent underwater acoustic (UWA) communications due to the low complexity of receivers required to deal with highly dispersive channels [1], [2]. Recently, deep learning (DL) has been applied successfully in OFDM channel estimation (CE) and signal detection over wireless channels [3], [4]. Unlike conventional receivers, DL can handle wireless channels in an end-to-end manner. It is widely acknowledged that the well-trained DL-receiver can not only reduce the receiver complexity [5], but also achieve perfect demodulation under unknown channels [6]. The design of DL-receivers can be generally classified into two categories. One is the data driven method, such as the classical FC-DNN [7], which takes into account the characteristics of data and the ability of neural networks, aiming to achieve global optimality. Whereas, most existing works based on the data-driven method consider the communication system as a black box. The other category is the model driven method, such as the well-known ComNet [8], which combines DL and expert knowledge. However, both of these two methods lack the prior structure of the OFDM receiver which makes DL-receivers sensitive to data. Moreover, most DL-receivers are

trained offline under dataset and deployed online in practical use, implying a difference between the training process and deploying process. This difference can seriously deteriorate the DL-receiver's performance. For example, FC-DNN and ComNet have poor generalization performance under mismatched channels [9]. In addition, it is very difficult to obtain the exact dataset in some communication scenarios, such as UWA communications, where the channel models may be unknown or mathematically intractable [6].

In this paper, we introduce multi-task learning (MTL) for UWA OFDM communications to improve the generalization performance. The MTL method has already achieved great success in computer vision, image segmentation and semantic recognition, which can improve the generalization performance of a neural network [10]. For classical OFDM receivers, the received signal is processed by fast Fourier transform (FFT), CE, equalization (EQ), and QAM demodulation (QD). To explore the strong relationship between these multiple tasks and exploit the prior structure of OFDM receiver, we apply the MTL to design a novel architecture for UWA OFDM receivers, called TaskNet, where the CE and EQ are considered as auxiliary tasks and the losses of auxiliary tasks are considered as penalty terms. Our main contributions are as follows.

- We create a baseband equivalent UWA channel dataset for DL by processing the UWA raw CIRs, which is called UWA-CIRset<sup>1</sup>, including simulated data for training and multi-scenario measured data for validation.
- We propose a novel OFDM receiver architecture based on MTL, namely TaskNet, by taking advantage of the sub-tasks to facilitate existing receiver models and the relationship between them. Tested on UWA-CIRset, the proposed TaskNet shows better generalization performance than the single-task methods, namely FC-DNN and ComNet. Even under mismatch conditions, the bit error rate (BER) performance of TaskNet is better than that of the traditional minimum mean square error (MMSE) algorithm and overwhelms that of FC-DNN and ComNet.

The rest of this letter is organized as follows. Section II provides a brief problem description for DL based receiver design. Then, the TaskNet architecture is introduced and analyzed in Section III. In Sections IV and V, a unified CIR dataset is created, and numerical results are demonstrated, respectively. Finally, conclusion is drawn in Section VI.

This work was supported by Key Program of Marine Economy Development (Six Marine Industries) Special Foundation of Department of Natural Resources of Guangdong Province (GDNRC [2020]009).

Corresponding author: Fei Ji; e-mail: eefeiji@scut.edu.cn.

<sup>1</sup>The dataset can be found from <https://github.com/haozhao10/Underwater-acoustic-channel-dataset> & <http://www2.scut.edu.cn/arcca/4621/list.htm>

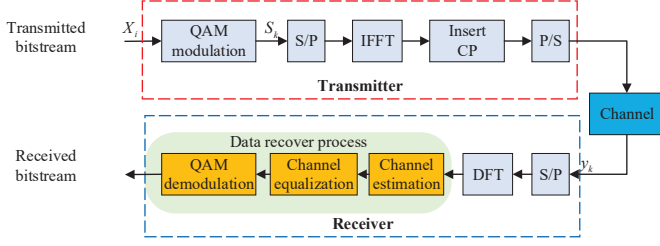


Fig. 1. The OFDM receiver structure.

## II. PROBLEM DESCRIPTION

The structure of the conventional OFDM communication system of  $N$  subcarriers is shown in Fig. 1. At the transmitter, the binary bit sequence  $\{X_1, \dots, X_i, \dots, X_{MN}\}$  are split into groups of  $M$  bits with  $i = 1, \dots, MN$ , and each group is mapped to one of the symbols of the  $2^M$ -QAM alphabet  $\mathcal{A}$ . The  $N$  generated symbols  $S_1, S_2, \dots, S_N$  are then fed to the inverse fast fourier transform (IFFT) to give the time-domain signal. After addition of a cyclic prefix (CP), the signal is transmitted over the channel.

At the receiver, after CP removal and FFT, the frequency domain received signal can be expressed as

$$Y_k = H_k S_k + W_k, \quad k = 1, \dots, N, \quad (1)$$

where  $Y_k$ ,  $H_k$ , and  $W_k$  denote the received signal, frequency channel response, and noise at the  $k$ -th subcarrier, respectively.

Subsequently, the received signal is processed serially by modules including CE, EQ, and QD. Taking the strong relationship between these modules into account, we can get a set of label  $T$  from offline simulations, which can be denoted as

$$T = \begin{Bmatrix} T^{CE}[\mathcal{R}(H_1, \dots, H_k, \dots, H_N), \mathcal{I}(H_1, \dots, H_k, \dots, H_N)] \\ T^{EQ}[\mathcal{R}(S_1, \dots, S_k, \dots, S_N), \mathcal{I}(S_1, \dots, S_k, \dots, S_N)] \\ T^{QD}[X_1, \dots, X_i, \dots, X_{MN}] \end{Bmatrix}, \quad (2)$$

where  $\mathcal{R}(\cdot)$  and  $\mathcal{I}(\cdot)$  refer to real part and imaginary part of the complex number.

## III. TASKNET

### A. Multi-Task Model Architecture

The architecture of the TaskNet receiver is shown in Fig. 2, containing data flow and loss flow, where **Con** refers to concatenation and **Cal** means calculation. The data flow works for forward calculation process and the loss flow is used for back propagation. Instead of using a black-box like FC-DNN to recover the received data, we associate every two layers with one task in the data flow. The CE and EQ subnets are initialed by traditional communication solutions, which is the same as the model-based data driven method [8].

In the CE task, the full connection layer without activation function is employed as

$$H_k^{ls} = \frac{Y_k}{P_k}, \quad (3)$$

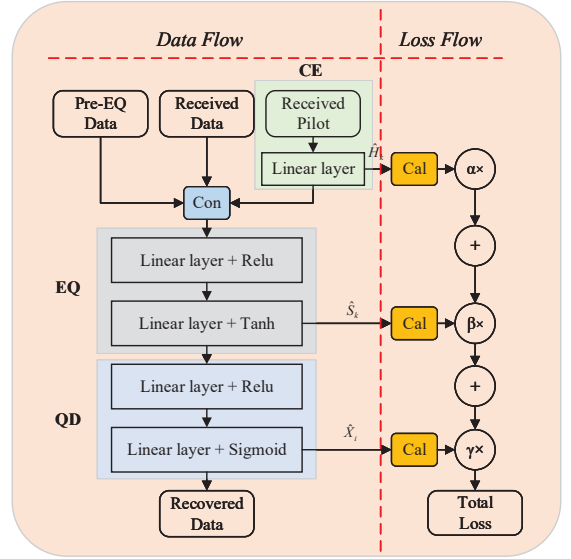


Fig. 2. The structure of Task-Net.

where  $P_k$  is the pilot symbol at the  $k$ th subcarrier. The CE task has no activation function, namely, it is a linear channel estimator which is recorded as

$$\hat{T}^{CE} = f^{CE}(H_k^{ls}). \quad (4)$$

The output of the CE task, received signal  $Y$  and pre-equalization serve as the input of the EQ subtask, i.e.,

$$I^{CE} = [\hat{T}^{CE}, \mathcal{R}(Y_k), \mathcal{I}(Y_k), \mathcal{R}(Y_k^{EQ}), \mathcal{I}(Y_k^{EQ})], \quad (5)$$

where we consider the pre-equalization based on the zero-forcing (ZF) as part of  $I^{CE}$  which can be obtained by

$$Y_k^{EQ} = \frac{Y_k}{H_k^{ls}}. \quad (6)$$

For the EQ task, the Relu function is used as the activation function in the input layer and the tanh activation function is applied to the output layer, which keeps the output within the interval  $[-1, 1]$ . Therefore, the EQ task can be expressed as

$$\hat{T}^{EQ} = f_{\tanh}^{EQ}(I^{CE}), \quad (7)$$

where

$$f_{\tanh}^{EQ}(x) = \frac{e^{\max(0, x)} - e^{-\max(0, x)}}{e^{\max(0, x)} + e^{-\max(0, x)}}. \quad (8)$$

The EQ task ensures that the received data can be mapped to constellation points.

After that, the QD task is employed to demap the constellation points to bits using the output of the EQ task. The sigmoid activation function is applied to the output layer, which limits the output within the interval  $[0, 1]$ . Thus, the QD task can be expressed as

$$\hat{T}^{QD} = f_{sigmoid}^{QD}(\hat{T}^{EQ}), \quad (9)$$

where

$$f_{sigmoid}^{QD}(x) = \frac{1}{1 + e^{-\max(0, x)}}. \quad (10)$$

From above, since some layers are shared by all tasks while the others are only related to specific tasks, we call it hierarchy MTL based receiver.

TABLE I  
PARAMETERS OF CHANNEL DATASET

| Parameters       | SIM-P     | SIM-B        | NOF    | NCS    | CWR                 |
|------------------|-----------|--------------|--------|--------|---------------------|
| Environment      | Rayleigh  | Default      | Fjord  | Shelf  | Reservoir           |
| Range            | -         | 500m~8000m   | 750m   | 540m   | 1100m, 2100m, 6000m |
| Water depth      | -         | 100m         | 10m    | 80m    | 50m                 |
| Transmitter depl | -         | Suspended    | Bottom | Bottom | Suspended           |
| Receiver depl    | -         | Suspended    | Bottom | Bottom | Suspended           |
| Doppler coverage | 30Hz      | Uncalculated | 7.8Hz  | 31.4Hz | Uncalculated        |
| Quantities       | 1,000,000 | 216,180      | 15,480 | 61,440 | 442,359             |

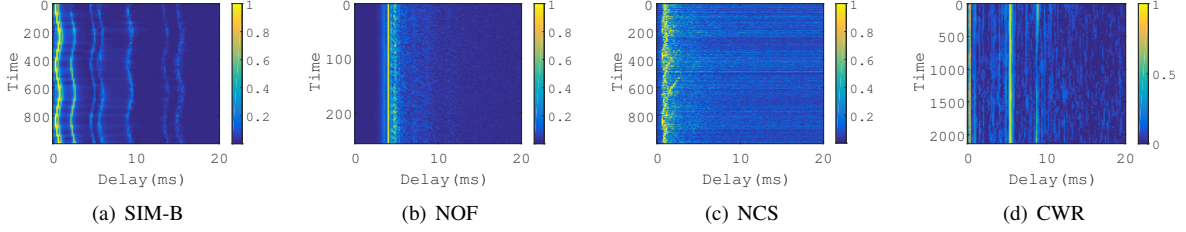


Fig. 3. A snapshot of CIR dataset.

### B. Loss Function Design

Usually, the single task receiver is trained by optimizing the loss function as

$$L_{STL} = L_B = \frac{1}{N} \sum_{i=1}^{MN} |T_i^{QD} - \hat{T}_i^{QD}|^2. \quad (11)$$

However, such loss function without any regularization makes the network sensitive to data. To avoid this issue, we design a MTL based loss function by utilizing the correlation between CE, EQ, and QD, and considering the CE and EQ tasks as auxiliary tasks. The loss function of the MTL based TaskNet can be formulated as

$$\begin{aligned} L_{MTL} &= \alpha L_B + \beta L_S + \gamma L_H \\ &= \frac{1}{N} \left( \alpha \sum_{i=1}^{MN} |T_i^{QD} - \hat{T}_i^{QD}|^2 + \beta \sum_{j=1}^{2N} |T_j^{EQ} - \hat{T}_j^{EQ}|^2 \right. \\ &\quad \left. + \gamma \sum_{j=1}^{2N} |T_j^{CE} - \hat{T}_j^{CE}|^2 \right), \end{aligned} \quad (12)$$

where  $L_S$  and  $L_H$  represent the losses of the EQ and QD tasks, respectively.  $\alpha, \beta, \gamma$  are weights of the three tasks to keep balance of each loss. The losses of CE and EQ tasks can be seen as a penalty term of (9). When CE and EQ performance is deteriorated, the penalty term will increase. In this way, TaskNet is prevented from overfitting on training dataset. Obviously, the performance of such system is strongly dependent on the relative weighting between each task.

### C. Training

In the cascaded MTL process, the whole network is trained in its entirety at each training step, followed by a random training batch. Then we update the model parameters by back-propagating the loss  $L_{MTL}$ . The auxiliary tasks are used as latent basis during the training process, which combining the losses of all tasks and minimizes the total loss. The initial learning rate value is 0.001.

## IV. UNIFIED CIR DATASETS

In this work, both simulated and measured CIRs are integrated into our dataset for a short range and shallow water communication scenario. The bandlimited UWA channel can be defined as an equivalent baseband response. The parameters of our dataset are shown in Table I. The channel sample rate is 6 kHz and the maximum spread delay is 20ms which corresponds to a channel length of 120 taps, containing all major multipaths. A snapshot of CIR is shown in Fig. 3. Since there is no precise channel simulation model for a variety of UWA communication scenarios [11]. It is meaningful to train a powerful DL-receivers using an imprecise training set.

### A. Training Dataset

In order to generate massive amounts of CIRs for learning, the probability model is selected to generate training set in this paper. We summarize the experience of our predecessors and create a training dataset of CIRs. The UWA multipath is simulated based on the probability model (SIM-P), which is rayleigh-distributed, and exponentially decays [12]. The UWA channel Doppler spread is considered by a bell-shaped function as

$$S(f) = \frac{\sqrt{a}}{\pi f_d \left[ 1 + a \left( \frac{f}{f_d} \right)^2 \right]}, \quad |f| \leq f_d, \quad (13)$$

where the maximum Doppler shift  $f_d$  is 50Hz and the coefficient  $a$  is 9.

### B. Validation Dataset

In this subsection, we provide a multi-scene validation dataset, containing simulation and measured CIRs with multi-communication environments, and multi-communication ranges. BELLHOP (SIM-B) is a widely-known UWA channel simulation method. Taking it into account, we create a part of data by [13]. The horizontal distance between the receiving

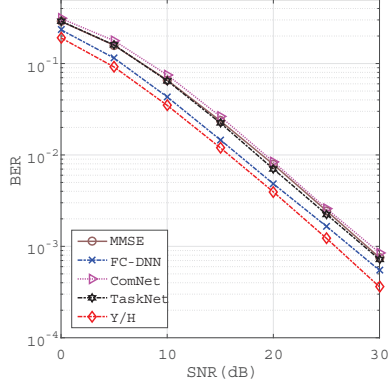


Fig. 4. BER curves of TaskNet and competing methods under SIM-P channel.

end and the transmitting end varies between 500m and 8000m, randomly. The vertical distance varies from 10m to 90m, randomly. In addition, the CIRs measured under different environments are considered. The raw CIRs were measured at Norway—Oslofjord (NOF), Norway—Continental Shelf (NCS) [14] and China—Wanlu Reservoir (CWR). It is worth noting that, in CWR, CIRs from different distances were collected. Both receiver and receiver terminals were located on two ships for long-distance communication test, with distances of 1100m, 2100m, and 6000m, respectively.

## V. SIMULATION RESULTS

In our experiments, Pytorch is chosen as development framework. The OFDM system with 512 sub-carriers and a CP of length 128 are considered. 4-QAM is used as the modulation scheme. The real and imaginary parts of 2 OFDM blocks containing the pilots and transmitted symbols are used as the input. Considering that the number of subcarriers is large while the sample data is limited [15], every 64 sub-carriers of an OFDM symbol are assigned to a group. We adopt the mean-squared error loss function and the adaptive moment estimation (Adam) optimizer. The following concise conventions are employed in the subsequent discussion.

**MMSE:** MMSE based receiver is used as classical algorithm for competition.

**FC-DNN:** FC-DNN in [7], as a representative of the data driven method.

**Y/H:** The quotient of received signal and full frequency domain information  $H$ , which can realize the maximum likelihood solution in the linear case.

**ComNet:** Proposed method in [8], which is a kind of model based data driven method, combining DL and expert knowledge in OFDM receivers.

**TaskNet:** The receiver architecture designed in this paper.

### A. BER Performance Under Matched Condition

Figure 4 compares the BER performance of TaskNet and the benchmark methods under matched channels, where all trained networks are tested over rayleigh fading channel. The networks of DNN and ComNet are rebuilt, and both are well-trained under our UWA data. From Fig. 4, FC-DNN, ComNet

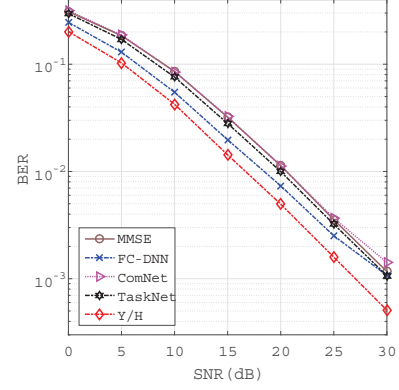


Fig. 5. BER curves of TaskNet and competing methods under clipping.

and TaskNet have similar performance under matched conditions. The well-known universal approximation theorem [17] enables the DL-receiver to achieve the best performance under matched condition. However, for UWA communication without precision training data, the BER performance under matched condition is unsatisfactory.

### B. Impact of Clipping and Filtering Distortion

In consideration of nonlinear characteristics in OFDM due to the high peak-to-average power ratio, the signal is clipped as in [7] with clipping ratio of 1. After clipping, nonlinear noise is introduced and the received data may be damaged. The clipped signal can expressed as

$$x(n) = \begin{cases} x(n), & \text{if } |x(n)| \leq A \\ Ae^{j\phi(n)}, & \text{otherwise} \end{cases}, \quad (14)$$

where  $A$  is the threshold of clipping,  $x(n)$  is the received signal and  $\phi(n)$  is the phase of  $x(n)$ . Figure 5 compares the BERs of TaskNet and the benchmark methods with clipping and filtering distortion. From Fig. 5, it is clear that, FC-DNN, ComNet and TaskNet can tolerate the nonlinear damage. However, in high SNR, TaskNet is more robust than FC-DNN and ComNet.

### C. BER Performance Under Mismatched Condition

To evaluate the DL-receiver, we test TaskNet, FC-DNN and ComNet in UWA-CIR data set. The CIR characteristics of all scenarios are totally different. NOF is a relatively benign communication channel while NCS is a tough channel. SIM-B pays more attention to the randomness of the scene. CWR provides data at different distances in the same scene. The BER curves of TaskNet and competing methods under NOF, NCS, SIM-B and CWR channels are shown in Fig. 6 (a)~(d), respectively. Judging from  $Y/H$ , we can find that the ranking of degree of fading in quasi-static is  $CWR < SIM-B < NOF < NCS$ . It is easy to see that whether the DL-receiver is tested under forgiving channels or tough channels, the performance of FC-DNN and ComNet shrink. That is because the mismatched condition makes the input of DL-receiver changing. FC-DNN and ComNet lack ability to

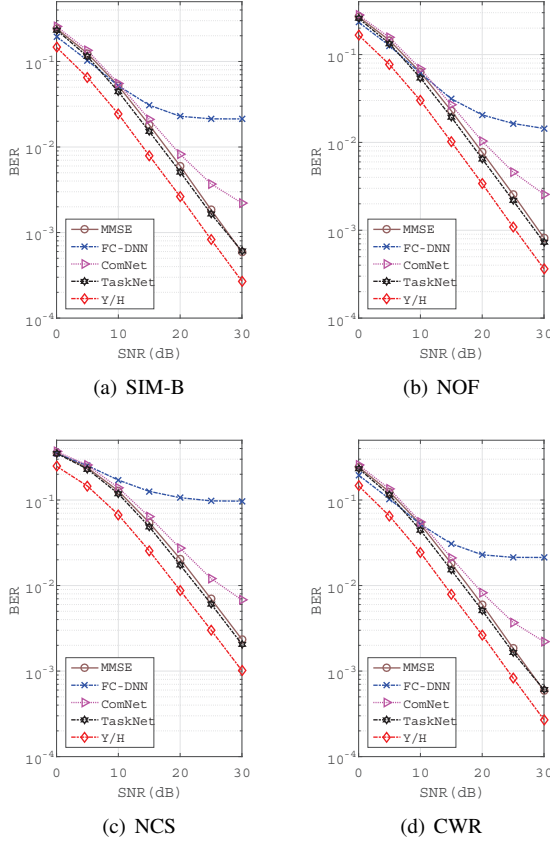


Fig. 6. BER curves of TaskNet and competing methods under different scenarios.

cope with this change. However, for TaskNet, we can see that the BER performance is still superior. In summary, the multi-task based TaskNet has better generalization performance than FC-DNN and ComNet both designed by a single task.

#### D. Complexity Analysis

Table II compares the complexities of the DL-receivers parameters including the amount of floating-point multiplication-adds (FLOPs), number of model parameters, model size and time consumption for processing a single-forward pass. All kinds of DL-receiver have same order time consumption on a single-forward process. The forward time is calculated on the CPU which is Intel®Core(TM)i5-4460 CPU @3.20GHz. From Table II, it is obvious that, TaskNet has tolerable complexity. It is worth noting that ComNet contains two separate training networks while TaskNet is trained as an integral. This method is in line with the original intention of end-to-end receiver design.

### VI. CONCLUSIONS

In this letter, we demonstrated the benefits of the proposed multi-task based receiver architecture to recover the transmit data in the UWA OFDM system under clipping damage condition and mismatch condition. Through the simulation results, the proposed TaskNet scheme was proved to have better generalization performance than FC-DNN and ComNet.

TABLE II  
COMPLEXITY ANALYSIS FOR TASKNET AND BENCHMARK METHODS.

| Name     | Flops    | Paras  | Memory | Integrated | Time (s)          |
|----------|----------|--------|--------|------------|-------------------|
| DNN      | 569.84M  | 0.285M | 1.09MB | YES        | $2.876 * 10^{-4}$ |
| Com-net  | 265.216M | 0.27M  | 1.01MB | NO         | $2.412 * 10^{-4}$ |
| Task-net | 589.824M | 0.29M  | 1.13MB | YES        | $3.155 * 10^{-4}$ |

From the perspective of MTL based DL assisted intelligence communication system, the useful penalty term can be created manually according to classical communication knowledge. Moreover, these features of related tasks can improve the generalization performance. The idea of taking advantage of the interplay between tasks as a way of regularizing inspires the future work of applying DLs to those scenarios with strong correlation.

### REFERENCES

- [1] B. Li, S. Zhou, M. Stojanovic, L. Freitag and P. Willett, "Multicarrier Communication Over Underwater Acoustic Channels With Nonuniform Doppler Shifts," *IEEE Journal of Oceanic Engineering*, vol. 33, no. 2, pp. 198-209, 2008.
- [2] M. Wen, X. Cheng, L. Yang, Y. Li, X. Cheng and F. Ji, "Index Modulated OFDM for Underwater Acoustic Communications," *IEEE Communications Magazine*, vol. 54, no. 5, pp. 132-137, 2016.
- [3] T. O'Shea and J. Hoydis, "An Introduction to Deep Learning for the Physical Layer," *IEEE Transactions on Cognitive Communications and Networking*, vol. 3, no. 4, pp. 563-575, Dec. 2017.
- [4] Z. Qin, H. Ye, G. Y. Li and B. F. Juang, "Deep Learning in Physical Layer Communications," *IEEE Wireless Communications*, vol. 26, no. 2, pp. 93-99, Apr. 2019.
- [5] E. Björnson and P. Giselsson, "Two Applications of Deep Learning in the Physical Layer of Communication Systems," 2020. [Online]. Available: arXiv: 2001.03350.
- [6] N. Farsad and A. Goldsmith, "Neural Network Detection of Data Sequences in Communication Systems," *IEEE Transactions on Signal Processing*, vol. 66, no. 21, pp. 5663-5678, Nov. 2018.
- [7] H. Ye, G. Y. Li and B. Juang, "Power of Deep Learning for Channel Estimation and Signal Detection in OFDM Systems," *IEEE Wireless Communications Letters*, vol. 7, no. 1, pp. 114-117, Feb. 2017.
- [8] X. Gao, S. Jin, C. Wen and G. Y. Li, "ComNet: Combination of Deep Learning and Expert Knowledge in OFDM Receivers," *IEEE Wireless Communications Letters*, vol. 7, no. 1, pp. 114-117, Feb. 2018.
- [9] P. W. Jiang, T. Q. Wang, B. Han, X. Gao, J. Zhang, C. K. Wen, S. Jin, and G. Y. Li, "Artificial intelligence-aided OFDM receiver: Design and experimental results," 2018. [Online]. Available: arXiv:1812.06638.
- [10] R. Caruana, "Multitask Learning," *Machine Learning*, vol. 28, no. 1, pp. 41-75, 1997.
- [11] A. Song, M. Stojanovic and M. Chitre, "Editorial Underwater Acoustic Communications: Where We Stand and What Is Next?" *IEEE Journal of Oceanic Engineering*, vol. 44, no. 1, pp. 1-6, 2019.
- [12] T. Ebihara and G. Leus, "Doppler-Resilient Orthogonal Signal-Division Multiplexing for Underwater Acoustic Communication," *IEEE Journal of Oceanic Engineering*, vol. 41, no. 2, pp. 408-427, Apr. 2016.
- [13] P. Qarabaqi and M. Stojanovic, "Statistical Characterization and Computationally Efficient Modeling of a Class of Underwater Acoustic Communication Channels," *IEEE Journal of Oceanic Engineering*, vol. 38, no. 4, pp. 701-717, 2013.
- [14] P. A. van Walree, F. Socheleau, R. Otnes and T. Jensenrud, "The Watermark Benchmark for Underwater Acoustic Modulation Schemes," *IEEE Journal of Oceanic Engineering*, vol. 42, no. 4, pp. 1007-1018, 2017.
- [15] K. Mei, J. Liu and X. Zhang, "Machine Learning Based Channel Estimation: A Computational Approach for Universal Channel Conditions," 2019. [Online]. Available: arXiv: 1911.03886.
- [16] Y. Zhang, J. Li, Y. Zakharov, X. Li and J. Li, "Deep Learning Based Underwater Acoustic OFDM Communications," *Applied Acoustics*, vol. 154, pp. 53-58, 2019.
- [17] K. Hornik, M. Stinchcombe and H. White, "Multilayer Feedforward Networks are Universal Approximators," *Neural networks*, vol. 2, no. 5, pp. 359-366, 1989.